Exercise 1:

a) We need to prove:

(1) ($B \in \mathcal{F}$): Let $A = B$. Then, by definition, $A \cap B = B \cap B = B \in \mathcal{F}$

(2) ($F \in \mathcal{F} \Rightarrow F^c \in \mathcal{F}$): Let $F \in \mathcal{F}$. Then there exists $A \in \mathcal{A}$ s.t $F = A \cap B$

Since $\mathcal{F}$ is σ-algebra in B:

$$F^c = B \setminus F = B \cap F^c = B \cap (A \cap B)^c = B \cap (A^c \cup B^c) = (B \cap A^c) \cup (B \cap B^c) = B \cap A^c$$

Since $A \in \mathcal{A}$ and $\mathcal{A}$ is σ-algebra, then $A^c \in \mathcal{A}$ and $A^c \cap B = F^c \in \mathcal{F}$

(3) ($F_n \in \mathcal{F}\ \forall n \in \mathbb{N} \Rightarrow \bigcup_{n \in \mathbb{N}} A_n \in \mathcal{F}$): By definition, $\exists_{A_n \in \mathcal{A}} \mid A_n \cap B = F_n\ \forall n \in \mathbb{N}$.

Then, $\bigcup_{n \in \mathbb{N}} F_n = \bigcup_{n \in \mathbb{N}} (A_n \cap B) = B \cap \bigcup_{n \in \mathbb{N}} A_n \in \mathcal{F}$ because $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{A}$ as $\mathcal{A}$ is σ-algebra. $\square$

If $B \in \Omega$ and $B \notin \mathcal{A}$, then property (2) and (3) hold as before, and we only need to prove property (1):

Let $A = \Omega$. Then, by definition, $A \cap B = \Omega \cap B = B \in \mathcal{F}$.

Therefore, the claim still holds: $\mathcal{F}$ is a σ-algebra in B.

b) We need to prove:

(1) ($\Omega \in \mathcal{C}$): By definition of $f: \Omega \to E$ and its preimage and since $E \in \mathcal{E}$ (as $\mathcal{E}$ is σ-algebra on E): $f^{-1}(E) = \{x \in \Omega : f(x) \in E\} = \Omega \in \mathcal{C}$.

(2) ($C \in \mathcal{C} \Rightarrow C^c \in \mathcal{C}$): Let $C \in \mathcal{C}$. Then, there exist $B \in \mathcal{E}$ such that $C = f^{-1}(B)$. Since B is σ-algebra, $B \in \mathcal{E} \Rightarrow B^c \in \mathcal{E}$ and then $f^{-1}(B^c) = (f^{-1}(B))^c = C^c \in \mathcal{C}$

(3) ($C_n \in \mathcal{C}\ \forall n \in \mathbb{N} \Rightarrow \bigcup_{n \in \mathbb{N}} C_n \in \mathcal{C}$). Let $C_n \in \mathcal{C}\ \forall n \in \mathbb{N}$, then there exist $B_n \in \mathcal{E}$ s.t. $C_n = f^{-1}(B_n)\ \forall n \in \mathbb{N}$. Since $\mathcal{E}$ is σ-algebra, then $\bigcup_{n \in \mathbb{N}} B_n \in \mathcal{E}$ and, by properties of the preimages,

$$f^{-1}\Big(\bigcup_{n \in \mathbb{N}} B_n\Big) = \bigcup_{n \in \mathbb{N}} f^{-1}(B_n) = \bigcup_{n \in \mathbb{N}} (C_n) \in \mathcal{C}. \quad \square$$